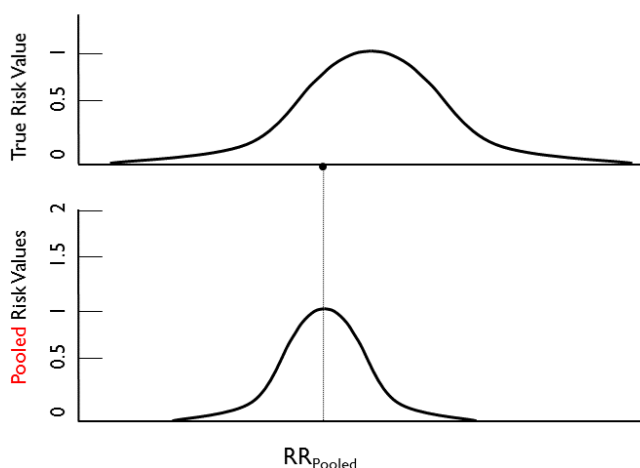


Pooling: Random-Effects



The Random-Effects model assigns each study a weight based on two factors:

1. The spread of estimates reported by that study (i.e. the variance)
2. How much that spread of estimates differs from the other studies

Finally, the Random-Effects model calculates a weighted average of the studies

Table 7-1 summarizes the different types of pooling approaches, and Appendix I provides a detailed discussion of the approaches. Note that some pooling methods are only available in Monte Carlo mode. This is because these pooling methods attempt to combine distributions of results into new distributions, and no distributional information is available in Point Mode. The Pooling Method column will thus have different values in its drop-down list depending on the mode used to generate the incidence results being pooled.

Table 7-1. Pooling Approaches for Incidence and Valuation Results

Pooling Approach	Description of Pooling Approach	Availability	
		Point Mode	Monte Carlo
None	No pooling performed.	Yes	Yes
Sum (Dependent)	Results are summed assuming they are perfectly correlated. In Point Mode, this is just a simple sum. In Monte Carlo mode, BenMAP-CE chooses the first point from each result in the pooling and does a simple sum to generate the first point in the pooled result, and so on for all of the points in the distribution of results.	Yes	Yes
Sum (Independent)	Results are summed assuming that they are independent. A Monte Carlo simulation is used. At each iteration, a random point is chosen from the distribution for each result, and the sum of these values is put in a holding container. After some number of iterations, the holding container is sorted low to high and binned down to the appropriate number of percentile points.	No	Yes

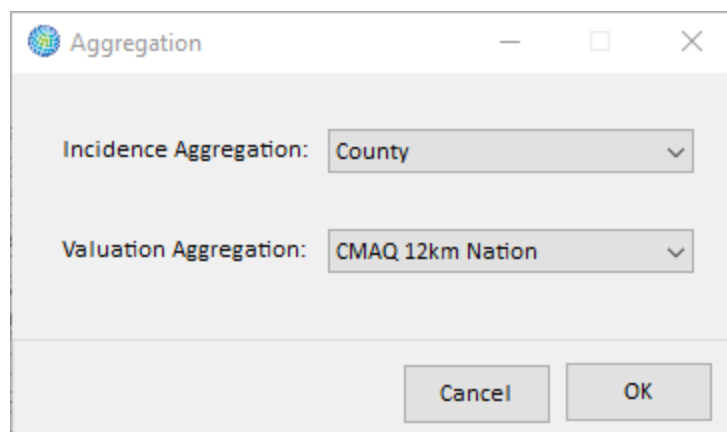
Pooling Approach	Description of Pooling Approach	Availability	
		Point Mode	Monte Carlo
Subtraction (Dependent)	Results are subtracted assuming they are perfectly correlated. All subsequent results are subtracted from the first result (the highest result in the display - to reorder results, simply click and hold a result and then drag it to its new position). In Point Mode, this is a simple subtraction. In Monte Carlo mode, BenMAP-CE chooses the first point from each result in the pooling and does a simple subtraction to generate the first point in the pooled result, and so on for all of the points in the distribution of results.	Yes	Yes
Subtraction (Independent)	Results are subtracted assuming that they are independent. A Monte Carlo simulation is used. At each iteration, a random point is chosen from the distribution for the first result, and then random points are chosen from the distribution for each subsequent result and subtracted from the first. The result is put into a holding container. After some number of iterations, the holding container is sorted low to high and binned down to the appropriate number of percentile points.	No	Yes
User defined Weights	Weights are specified by the user. In Point Mode, the new result is generated by a simple weighted sum of the input results. In Monte Carlo mode, the results are combined using the user specified weights with the 'Round Weights to Two Digits' Advanced Pooling Method. Note that the weights you enter need not add up to one - BenMAP-CE will normalize them internally. Also note that BenMAP-CE initializes all the weights to $1/n$, where n is the number of results being pooled.	Yes	Yes

7.2 Create Aggregation, Pooling, and Valuation (APV) Configuration

Once you have run a configuration result file (see Chapter 6), you can begin creating your APV configuration. You will start with selecting the aggregation levels for incidence and valuation results, then move on to pooling and valuation. These processes are described in detail below.

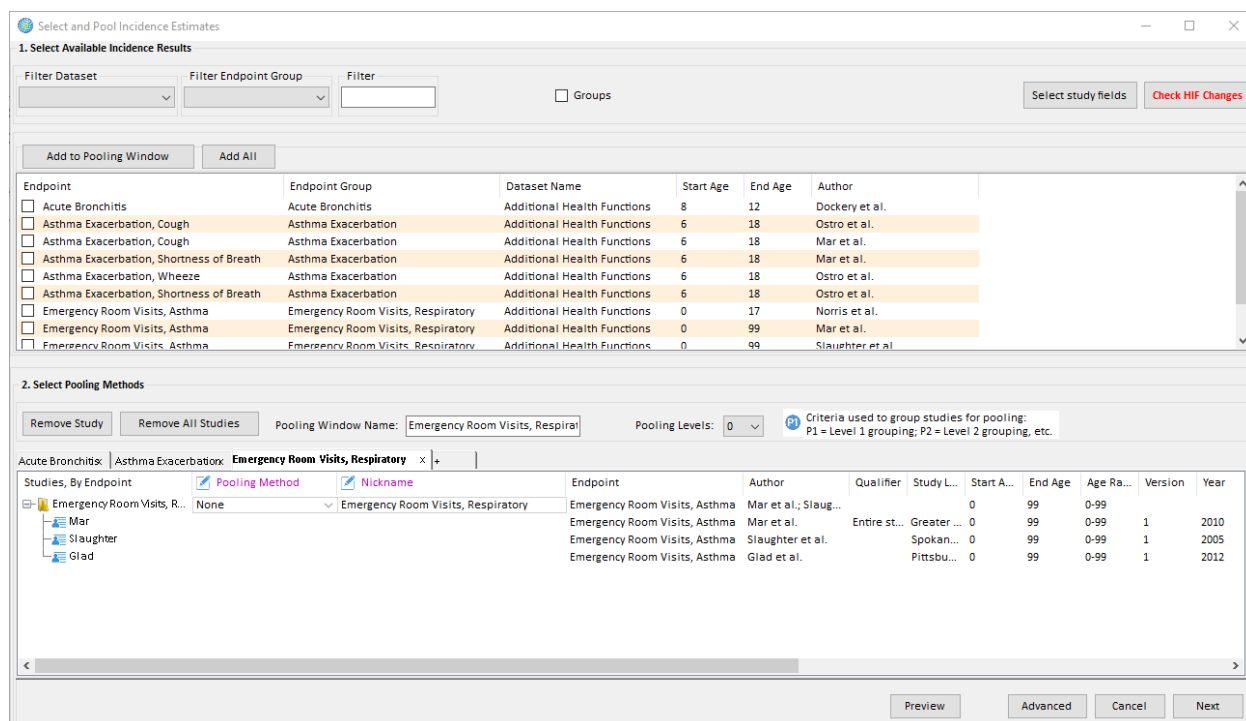
7.2.1 Selecting Aggregation Levels

Double-clicking **Aggregation** from the BenMAP-CE tree menu item opens the window which lets you choose the level of aggregation for the incidence and valuation results.



7.2.2 Pooling Incidence Results

To begin pooling incidence results, double-click **Pooling Method** from the BenMAP-CE tree menu. In the top half of the **Select and Pool Incidence Estimates** window, you will find a list of **Available Incidence Results**. The results are represented by the health impact functions from which they were created. You may click on the **Select study fields** button to display additional information about each health impact function. This is recommended when using the geographic areas functionality (i.e., applying region-specific air pollution effect coefficients).



There are several steps to pooling your incidence results:

Step 1. Add incidence results to the pooling window

Using the available incidence results, you can drag individual incidence results to the pooling window, or select the result(s) using the checkboxes next to each study, and click **Add to Pooling Window** to move them to the pooling window. You do not have to add all of your incidence results to the pooling window, but note that only those results showing in the pooling window will be included in the pooled incidence or valuation results. As with health impact functions, in Chapter 6, there is a filter method below the **Available Incidences Results** for your convenience.

The bottom half of the pooling window consists of one or more tabs, each with an associated “pooling window”, where health impact functions can be added and specifications for how to pool them (if any) can be made. In version 1.5, by default, BenMAP-CE will automatically create a pooling window for each Endpoint Group for which one or more Health Impact Functions were selected in the Health Impact Function selection screen, and the Endpoint Group will be indicated on the tab. For example, there could be one pooling window for Mortality and one for Asthma Exacerbation. Health impact functions should be added to the pooling window corresponding to that function’s Endpoint Group. Additional pooling windows can be added if needed, as discussed below.

Incidence results are displayed in the pooling window in a tree structure determined by (1) the order of the columns, and (2) the values of the identifying variables of the **Health Impact Functions** from which the incidence results were generated (**Endpoint Group**, **Endpoint**, etc.).

BenMAP-CE Best Practices

Incidence results *must* be added to the pooling window in order to be included in the valuation step, whether or not you intend to pool them with other incidence results. If you would like to apply a valuation function to a single (unpooled) health incidence estimate, add the health incidence estimate to the pooling window and specify the **Pooling Method** as “None.”

BenMAP-CE Guidance and Best Practices

BenMAP-CE automatically generates a pooling window (i.e., tab) for each unique **Endpoint Group** present in your health incidence results.

Generally, you should add health impact functions to the pooling window corresponding to that function’s **Endpoint Group**. For example, you would likely want to add health incidence results for *Mortality, Respiratory* to the “Mortality” pooling window.

You may need multiple pooling windows for the same **Endpoint Group** if you want to pool and/or value the same incidence results in different ways. For example, you might have age-stratified incidence results and want to calculate age-specific pooled results as well as a pooled result across all ages. In this case, you would need to set up two pooling windows for the same **Endpoint Group**. You may use as many pooling windows as you need in your analysis; you should give each pooling window a unique name. See Step 4 for more information on adding pooling windows.